

DATA ANALYTICS PROJECT

SOCIAL MEDIA ADDICTION AND MENTAL HEALTH ANALYSIS

Using MySQL and PowerBI

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INTRODUCTION TO MYSQL AND POWERBI

MySQL

- MySQL is a relational database management system.
- It is used to store, manage, and query structured data.
- In this project, MySQL was used to perform analytical queries on the dataset.

PowerBI

- Power BI is a business intelligence and data visualization tool.
- It helps convert raw data into dashboards and interactive visuals.
- In this project, Power BI was used to visualize addiction and mental health patterns.

PROJECT OVERVIEW

➔ The dataset contains:

- Student ID
- Age
- Gender
- Country
- Academic Level
- Daily Usage Hours
- Sleep Hours
- Addiction Score
- Mental Health Score
- Social Media Conflicts
- Academic Impact
- Most Used Platform

➔ Aim: To analyze the relationship between social media usage, addiction, sleep patterns, mental health, and academic performance among students.

COUNTRIES WITH THE HIGHEST AVERAGE ADDICTION SCORE

MySQL Query and Table

```
select Country, round(avg(Addicted_Score),2)
as Avg_Addiction_Score
from addiction
group by Country
order by Avg_Addiction_Score desc
limit 10;
```

Country	Avg_Addiction_Score
Lebanon	9.00
Czech Republic	9.00
Liechtenstein	9.00
Armenia	9.00
Ecuador	9.00
USA	8.60
UAE	8.13
Andorra	8.00
Thailand	8.00
Bhutan	8.00

Insight

- Lebanon, Czech Republic, Liechtenstein, Armenia, and Ecuador recorded the **highest average addiction score of 9.0**.
- The USA followed with an **average addiction score of 8.6**.
- Several countries showed similarly high scores, suggesting social media addiction is a **global concern** rather than being limited to a specific region.

QUESTIONS

- Which social media platforms are used most frequently?
- Is higher social media usage associated with higher addiction?
- How is addiction related to mental health?
- Is sleep duration related to mental well-being?
- Are social media conflicts linked to addiction?
- Does social media impact academic performance?
- Is there a relationship between usage hours and sleep duration?

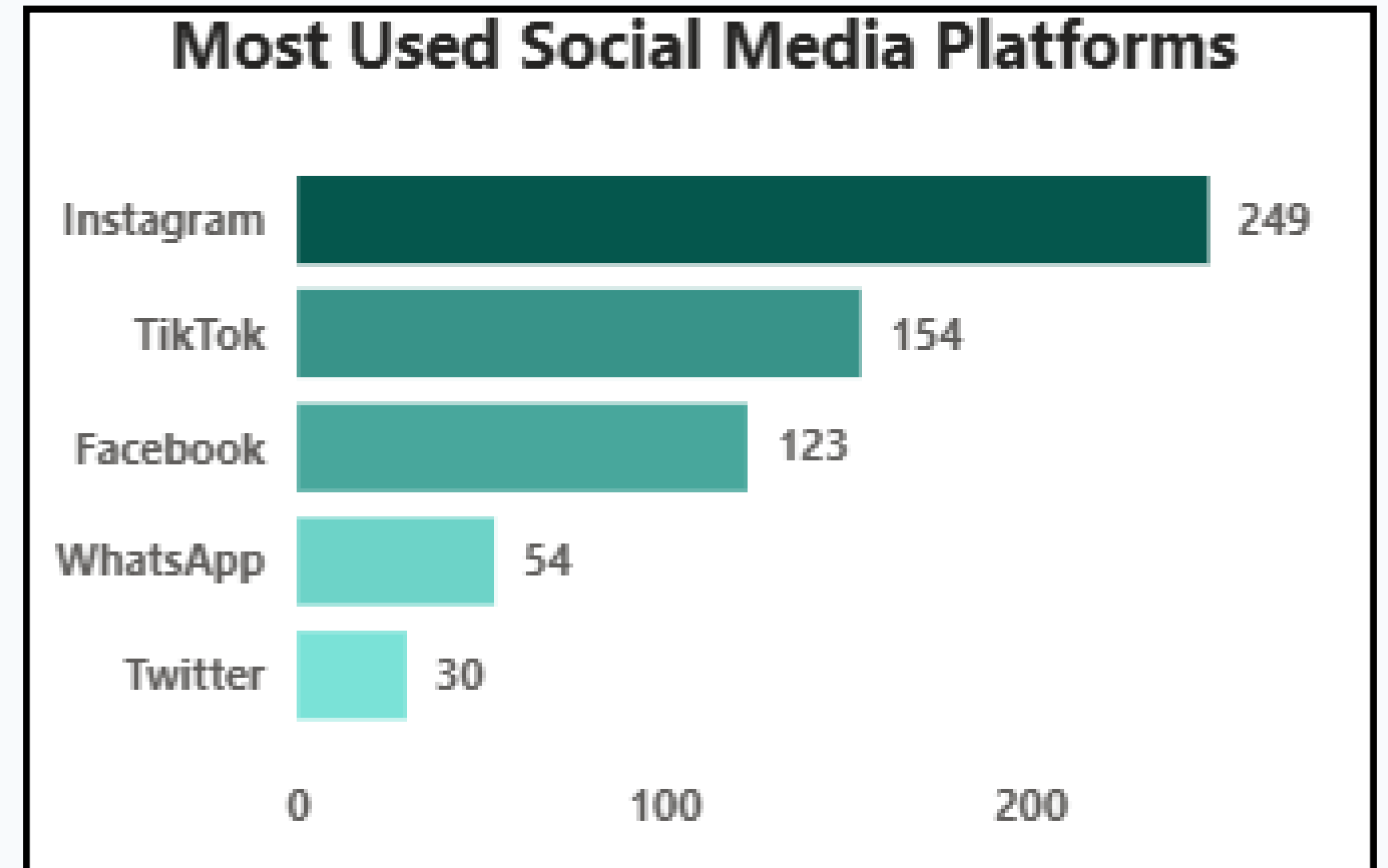
1. WHICH PLATFORMS ARE USED MOST FREQUENTLY?

MySQL Query and Table

```
select Most_Used_Platform, count(*) as  
No_of_Students  
from addiction  
group by Most_Used_Platform  
order by No_of_Students desc;
```

Most_Used_Platform	No_of_Students
Instagram	249
TikTok	154
Facebook	123
WhatsApp	54
Twitter	30
LinkedIn	21
WeChat	15
Snapchat	13
LINE	12
KakaoTalk	12
VKontakte	12
YouTube	10

PowerBI Visual



Instagram is the most frequently used platform among students.

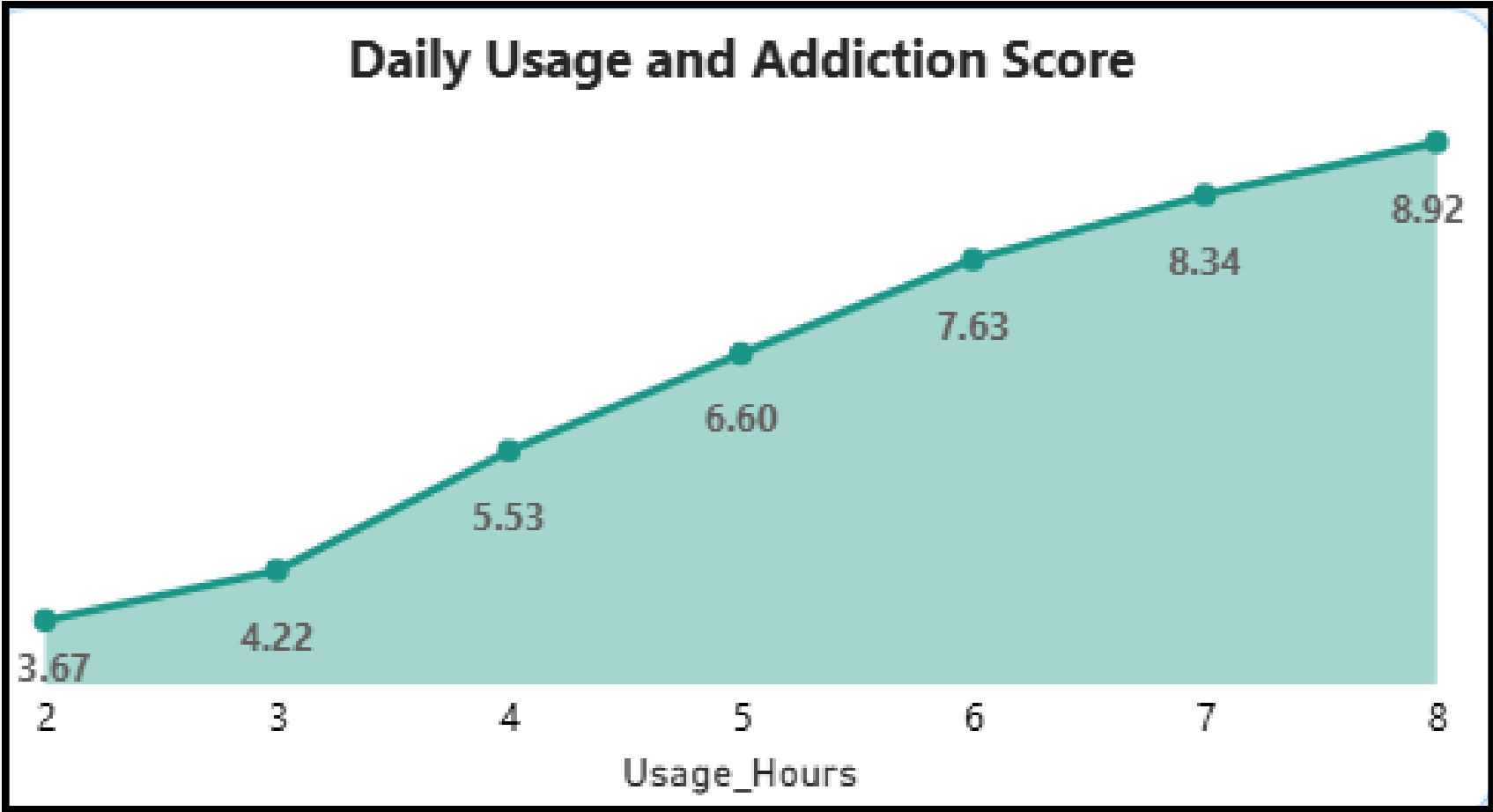
2. DOES HIGHER USAGE INCREASE ADDICTION?

MySQL Query and Table

```
select round(Avg_Daily_Usage_Hours) as
Usage_Hours, round(avg(Addicted_Score),2)
as Avg_Addiction
from addiction
group by round(Avg_Daily_Usage_Hours)
order by Usage_Hours;
```

Usage_Hours	Avg_Addiction
2	3.67
3	4.22
4	5.53
5	6.60
6	7.63
7	8.34
8	8.92

PowerBI Visual



Addiction score increases steadily as daily usage rises from 2 to 8 hours.

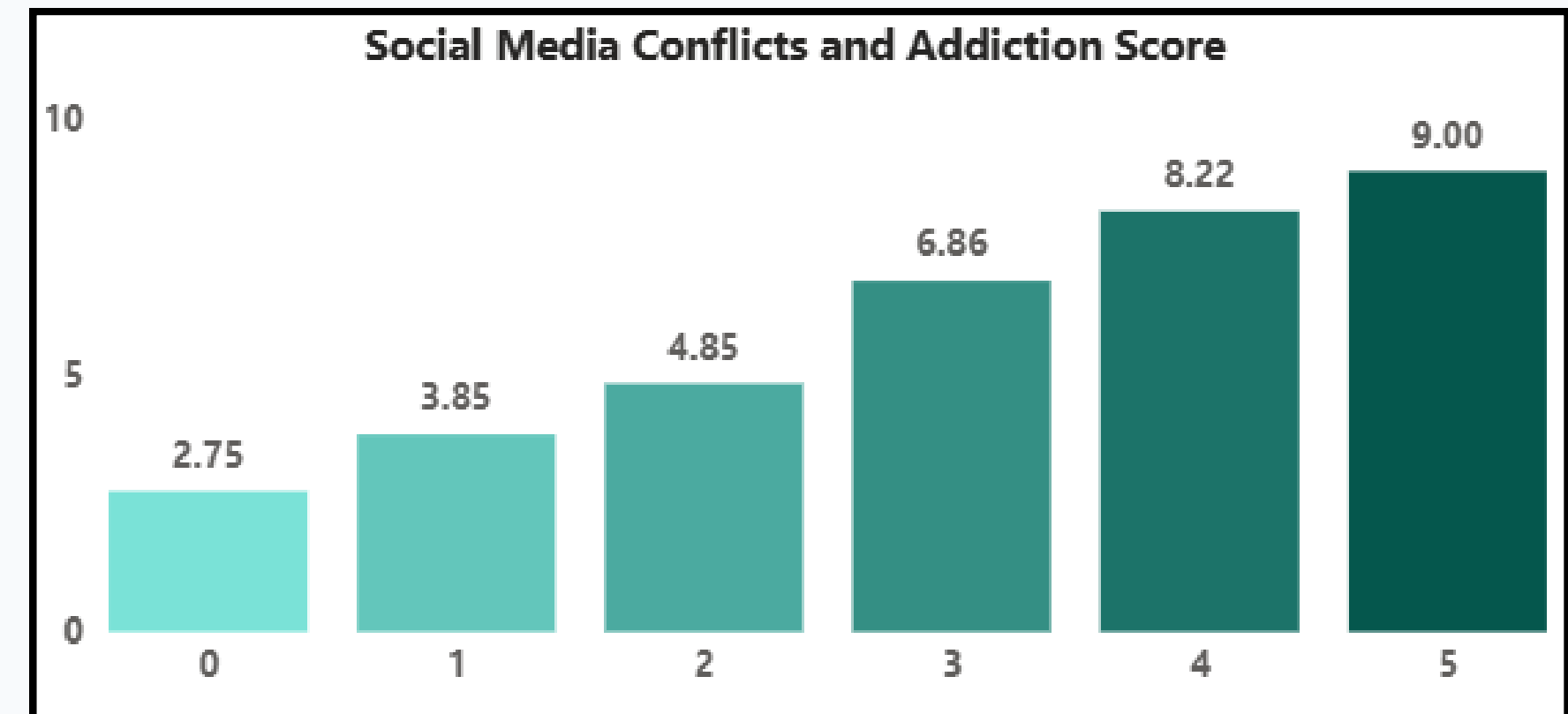
3. ARE SOCIAL MEDIA CONFLICTS RELATED TO ADDICTION?

MySQL Query and Table

```
select Conflicts_Over_Social_Media,  
round(avg(Addicted_Score),2) as Avg_Addiction  
from addiction  
group by Conflicts_Over_Social_Media  
order by Conflicts_Over_Social_Media;
```

Conflicts_Over_Social_Media	Avg_Addiction
0	2.75
1	3.85
2	4.85
3	6.86
4	8.22
5	9.00

PowerBI Visual



Students reporting more social media conflicts exhibit significantly higher addiction scores.

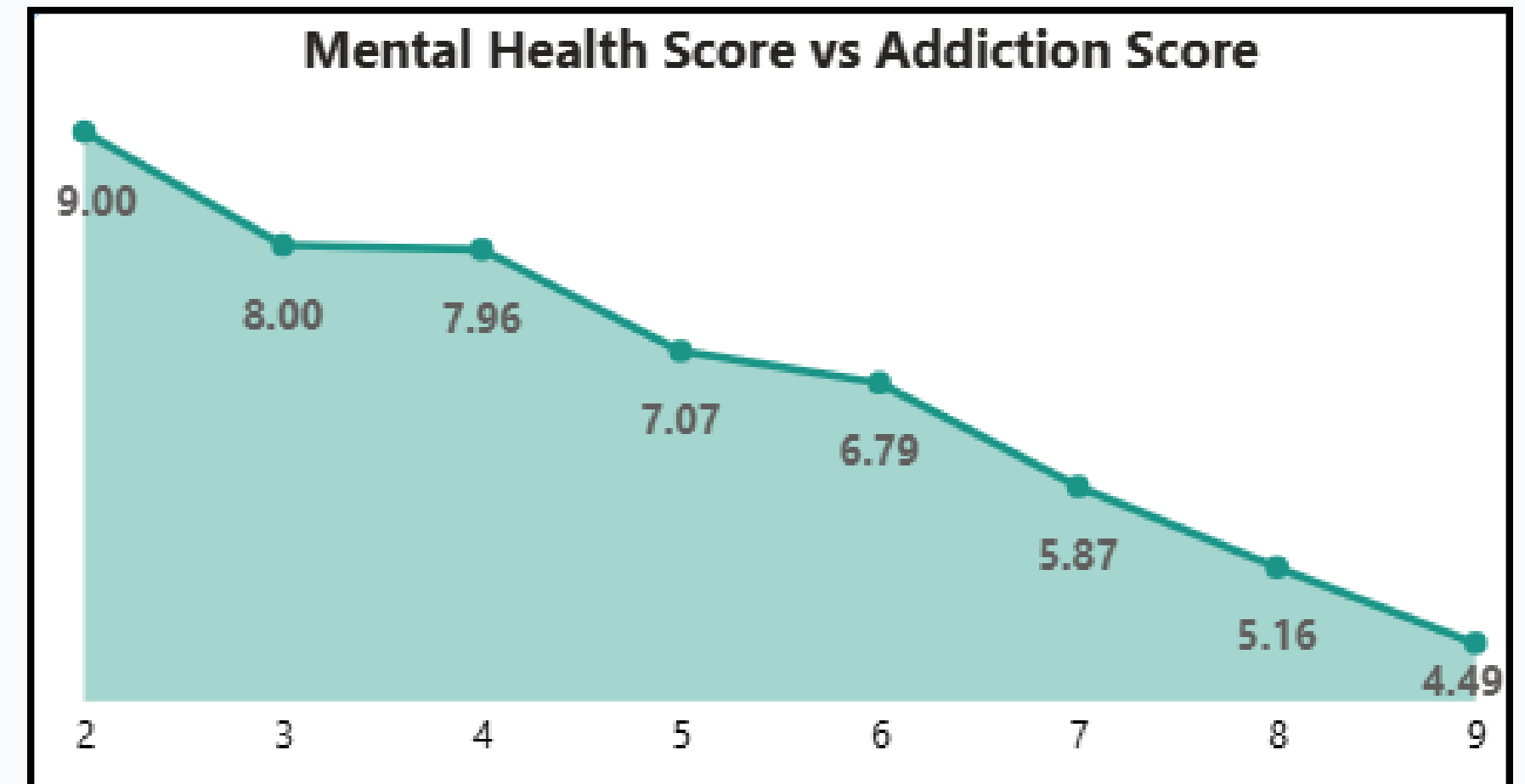
4. IS ADDICTION RELATED TO MENTAL HEALTH?

MySQL Query and Table

```
select Addicted_Score,  
round(avg(Mental_Health_Score),2) as  
Avg_Mental_Health  
from addiction  
group by Addicted_Score  
order by Addicted_Score;
```

Addicted_Score	Avg_Mental_Health
2	9.00
3	8.00
4	7.96
5	7.07
6	6.79
7	5.87
8	5.16
9	4.49

PowerBI Visual



Mental health scores decline consistently as addiction scores increase.

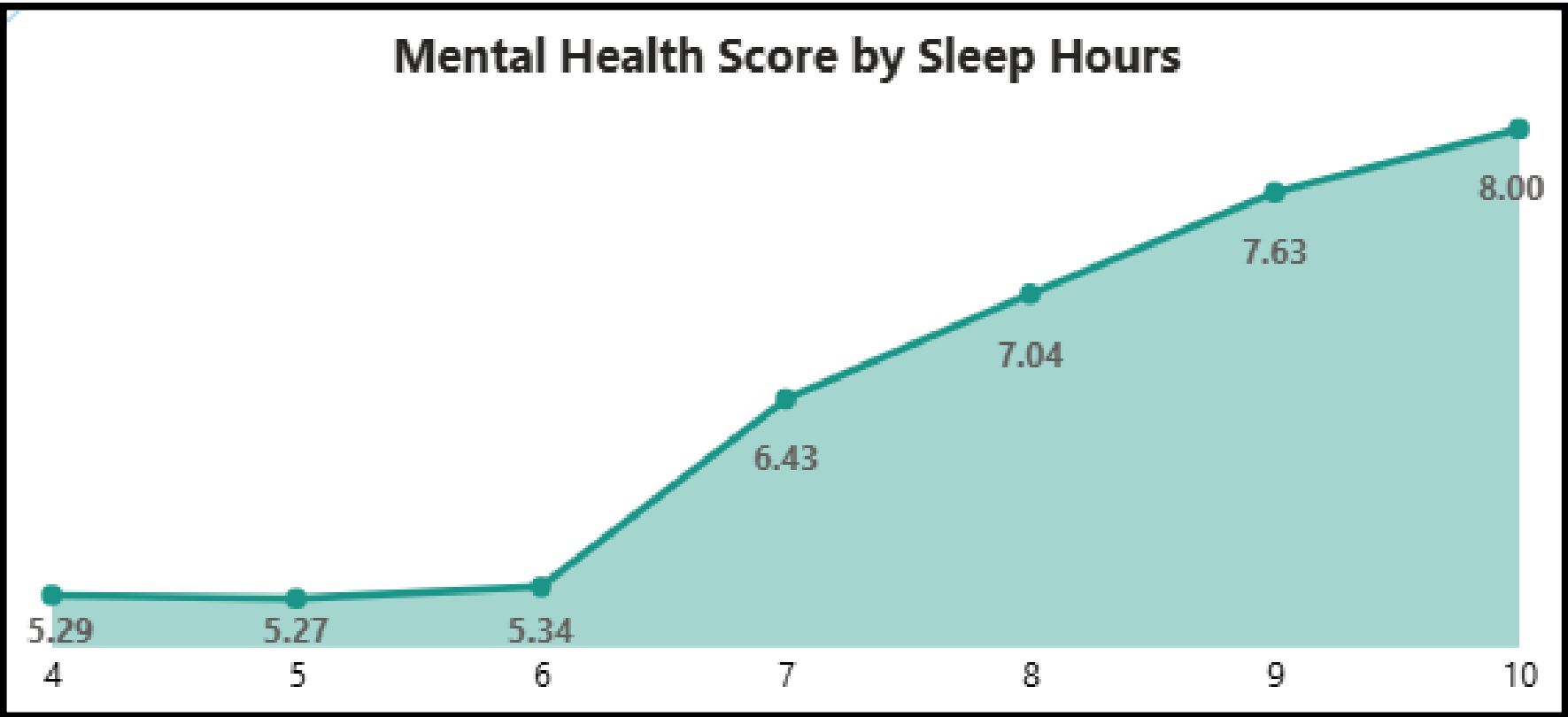
5. IS SLEEP DURATION RELATED TO MENTAL HEALTH?

MySQL Query and Table

```
select round(Sleep_Hours_Per_Night) as  
Sleep_Hours, round(avg(Mental_Health_Score),2)  
as Avg_Mental_Health  
from addiction  
group by round(Sleep_Hours_Per_Night)  
order by Sleep_Hours;
```

Sleep_Hours	Avg_Mental_Health
4	5.29
5	5.27
6	5.34
7	6.43
8	7.04
9	7.63
10	8.00

PowerBI Visual



Students sleeping 9–10 hours report the highest mental health scores.

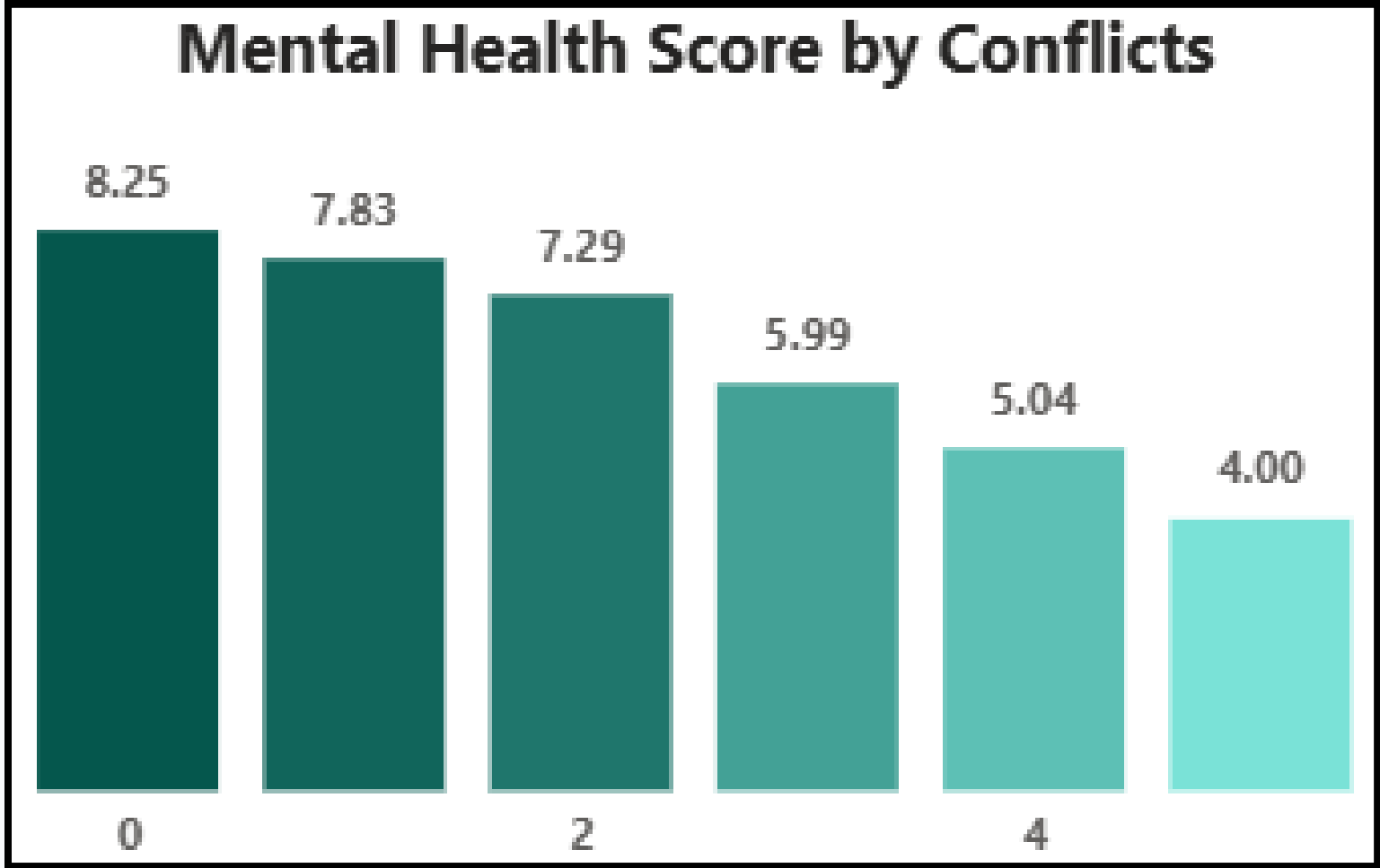
6. IS MENTAL HEALTH RELATED TO SOCIAL MEDIA CONFLICTS?

MySQL Query and Table

```
select Conflicts_Over_Social_Media,  
round(avg(Mental_Health_Score),2) as  
Avg_Mental_Health  
from addiction  
group by Conflicts_Over_Social_Media  
order by Conflicts_Over_Social_Media;
```

Conflicts_Over_Social_Media	Avg_Mental_Health
0	8.25
1	7.83
2	7.29
3	5.99
4	5.04
5	4.00

PowerBI Visual



Mental health scores decrease as conflict frequency increases.

7. DOES SOCIAL MEDIA AFFECT ACADEMICS?

MySQL Query and Table

```
select Affects_Academic_Performance,  
round(avg(Addicted_Score),2) as Avg_Addiction  
from addiction  
group by Affects_Academic_Performance;
```

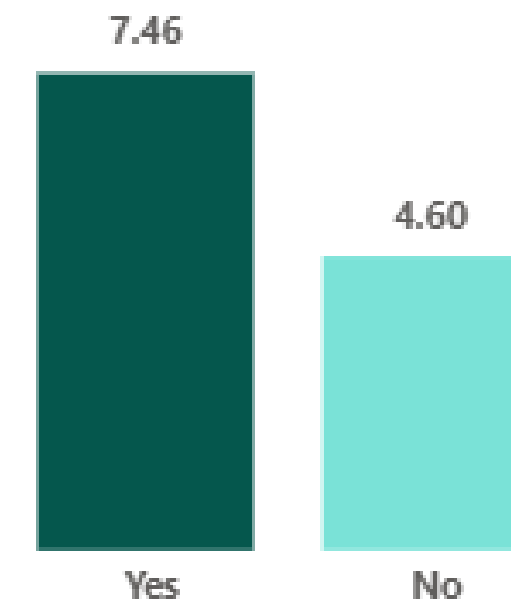
Affects_Academic_Performance	Avg_Addiction
Yes	7.46
No	4.60

```
select Affects_Academic_Performance,  
round(avg(Sleep_Hours_Per_Night),2) as Avg_Sleep  
from addiction  
group by Affects_Academic_Performance;
```

Affects_Academic_Performance	Avg_Sleep
Yes	6.34
No	7.81

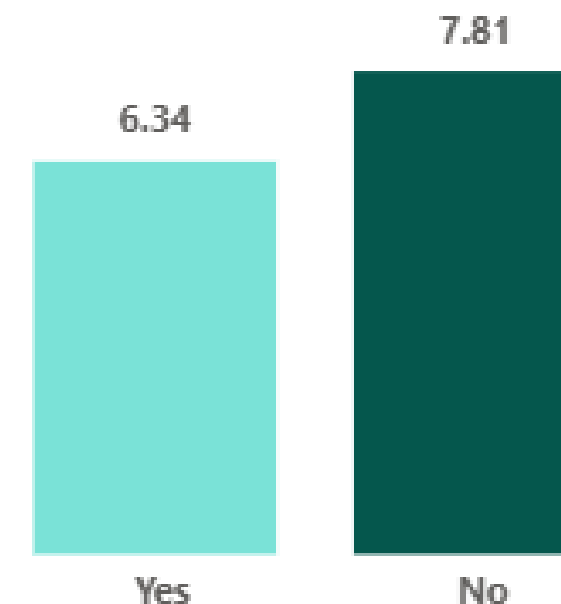
PowerBI Visual

Academic Impact vs Addiction Score



Students who reported academic impact had higher addiction scores (7.46 vs 4.60).

Sleep Hours by Academic Impact



Students who reported academic impact had lower average sleep duration (6.34 vs 7.81 hrs).

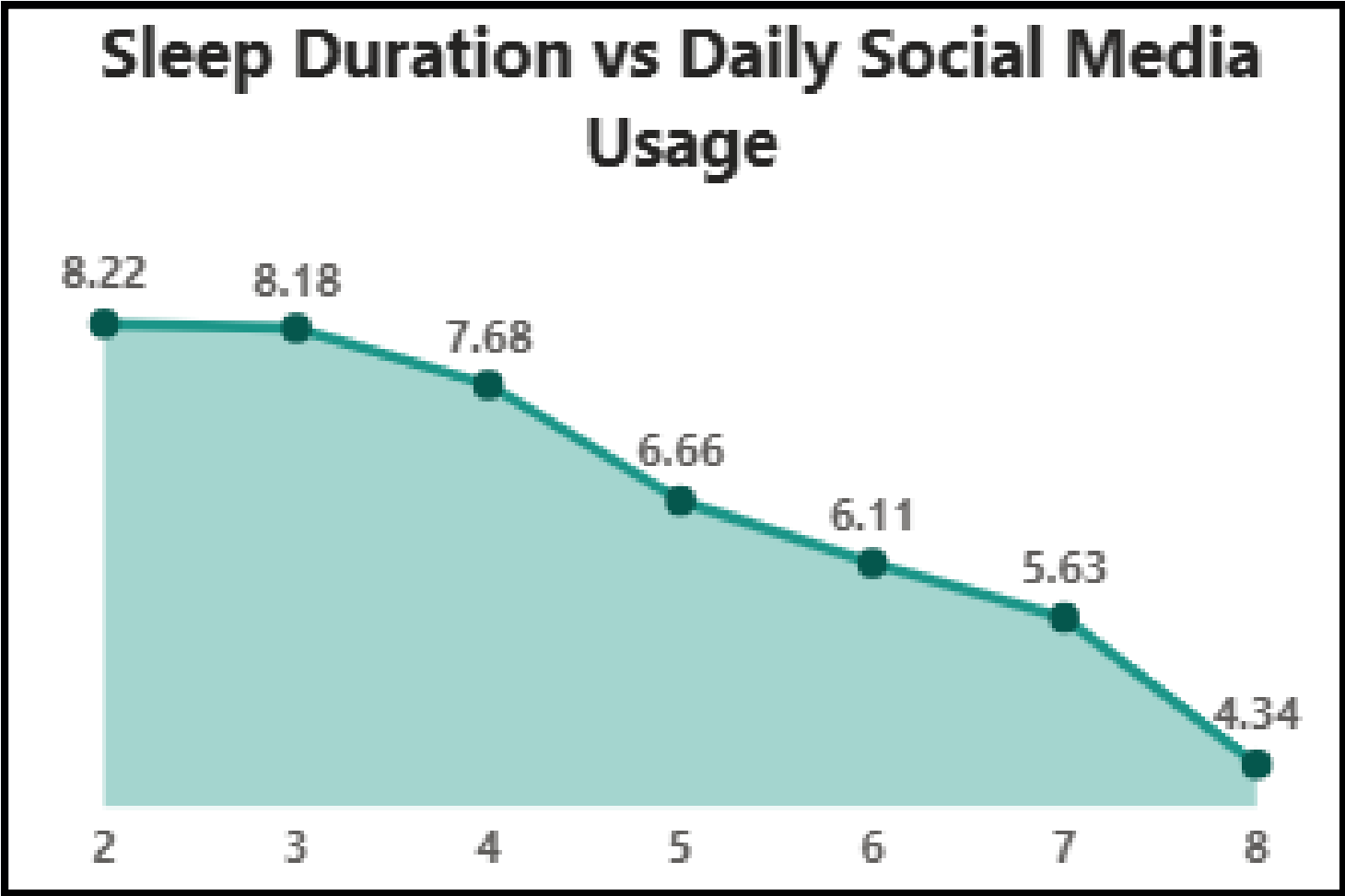
8. IS HIGHER USAGE ASSOCIATED WITH LOWER SLEEP DURATION?

MySQL Query and Table

```
select round(Avg_Daily_Usage_Hours) as
Usage_Hours, round(avg(Sleep_Hours_Per_Night),2)
as Avg_Sleep
from addiction
group by round(Avg_Daily_Usage_Hours)
order by Usage_Hours;
```

Usage_Hours	Avg_Sleep
2	8.22
3	8.18
4	7.68
5	6.66
6	6.11
7	5.63
8	4.34

PowerBI Visual



Average sleep duration decreases from 8.22 hours to 4.34 hours as social media usage increases.

KEY INSIGHTS FROM THE PROJECT

- **Instagram** is the **most used platform** among students.
- **Addiction scores rise** significantly with **increased social media usage**.
- **Mental health declines** as **addiction increases**.
- **More social media conflicts** are associated with **lower mental health**.
- Students **sleeping 9–10 hours** report the **highest mental health scores**.
- **Sleep duration drops** sharply as **daily usage increases**.
- Students reporting **academic impact** exhibit **higher addiction scores** and **lower sleep duration**.

SUGGESTIONS

- **Limit daily social media usage to reduce addiction risk.**
- **Encourage adequate sleep duration to support mental well-being.**
- **Promote awareness about the impact of excessive social media use on mental health.**
- **Encourage students to manage online conflicts effectively.**
- **Introduce digital well-being programs to support healthy social media habits.**

CONCLUSION

- **MySQL was used to analyze student social media behavior through SQL queries.**
- **Power BI transformed the data into interactive dashboards.**
- **Balanced social media usage and healthy sleep habits are associated with better well-being and academic outcomes.**

THANK YOU!